Timesheet Management System (TMS)

Software Requirement Specification (SRS)

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1. Project Overview

The objective of this project is to develop a centralized, web-based **Timesheet Management System (TMS)** to replace the current Excel-based manual resource utilization tracking process.

The application will allow employees/resources to log daily or weekly working hours against assigned projects while enabling managers and administrators to monitor project utilization, employee allocation, and effort distribution through dashboards and reports.

The system must support:

- Multi-project resource allocation
- Daily and weekly hour tracking
- Role-based access control
- Automated validations
- Reporting and analytics
- Secure and scalable web architecture

2. Business Objectives

The proposed system aims to:

- Eliminate manual Excel tracking
- Improve accuracy of time utilization
- Prevent over-utilization of resources
- Provide centralized reporting
- Increase visibility into project effort allocation
- Improve resource planning and management
- Enable management-level utilization dashboards

3. Application Scope

The application will be developed as a responsive web-based platform accessible through browsers.

The system will support:

- Employees/Resources
- Managers
- Administrators

The application will manage:

- Resources
- Projects
- Business Units
- Timesheets
- Utilization Reports

4. User Roles & Permissions

A. Admin

The Admin will have complete access to the system.

Responsibilities:

- Create/Edit/Delete Resources
- Create/Edit/Delete Projects
- Assign projects to resources
- Configure business units
- View all timesheets
- Generate reports
- Manage user roles and permissions

B. Manager

Managers will have access to their team members and assigned projects only.

Responsibilities:

- View team timesheets
- Approve/reject submitted timesheets
- View utilization dashboards
- Generate project-level reports

C. Employee/Resource

Resources can access only their assigned projects and personal timesheets.

Responsibilities:

- Submit timesheets
- Edit draft timesheets
- View own utilization
- View assigned projects only

5. Functional Requirements

5.1 Resource Management

The system must allow Admin users to:

- Add new resources/employees
- Update employee details
- Assign managers
- Assign projects
- Activate/deactivate users

Resource Fields

- Employee ID

- Employee Name
- Email ID
- Designation/Role
- Department/Business Unit
- Reporting Manager
- Status (Active/Inactive)

5.2 Project Management

The system must allow Admin users to manage projects.

Project Fields

- Project ID
- Project Name
- Business Unit
- Project Manager
- Start Date
- End Date
- Status

5.3 Resource Allocation

The system must support assigning multiple projects to a single resource.

Key Rules

- One resource can work on multiple projects
- Allocation visibility must be role-based
- Employees should only view assigned projects

5.4 Timesheet Entry Module

Resources should be able to log effort against projects.

Timesheet Features

- Daily or weekly time entry
- Project-wise hour allocation
- Month selection
- Week-wise breakdown
- Auto-save draft functionality
- Submit for approval

5.5 Validation Rules

Daily Hour Validation

The total logged hours for a resource must NOT exceed:

$$\text{Total Daily Hours} \leq 8$$

System Behavior

- If total hours exceed 8 hours:
    - Show validation error
    - Prevent submission/save

Weekend Rules

- **We can create Holiday Project. The admin can create holiday project based on regional and geographical holidays.**
- Saturdays and Sundays should be excluded from standard working calculations
- Optional future enhancement:
    - Allow weekend entries with approval

Project Visibility Rule

- Resources should only view:
    - Assigned projects
    - Their own submitted timesheets

5.6 Timesheet Approval Workflow

Workflow Steps

1. Resource submits timesheet
2. Manager reviews submission
3. Manager approves/rejects
4. Resource receives notification

Statuses

- Draft
- Submitted
- Approved
- Rejected

5.7 Dashboard & Reporting

The system should provide dashboard capabilities similar to Excel Pivot Tables. The dashboard should specific to each role like Admin can access the full dashboard. Manager can see his/her team utlilastion etc.

Dashboard Features

- Resource utilization dashboard
- Project effort summary
- Business Unit utlilization
- Monthly effort trends

Filters

- Month/Year

- Resource
- Project
- Business Unit
- Manager

Reports

- Resource-wise hours
- Project-wise utilization
- Monthly utilization
- Under-utilized resources
- Over-utilized resources

5.8 Notifications

The system should support:

- Timesheet submission reminders
- Approval notifications
- Rejection notifications

(Optional future enhancement: Email integration)

6. Non-Functional Requirements

Performance

- Application should support concurrent users efficiently
- Dashboard reports should load within acceptable response time

Security

- Role-based access control (RBAC)
- Secure authentication

- Encrypted passwords
- Session management

Scalability

The application should support future enhancements including:

- Leave management
- Attendance integration
- Payroll integration
- Mobile application support

Usability

- Responsive UI
- Modern user-friendly design
- Easy navigation
- Minimal training required

7. Suggested Technology Stack

Frontend

- Angular / React / Next.js

Backend

- .NET Core Web API / Node.js

Database

- SQL Server / PostgreSQL

Authentication

- JWT Authentication
- Azure AD (Optional Future Enhancement)

Hosting

- Azure App Service / IIS

8. Database Entities

Core Tables

- Users
- Roles
- Projects
- BusinessUnits
- ResourceProjectMapping
- Timesheets
- TimesheetEntries
- ApprovalHistory

9. Future Enhancements

- Mobile application
- Integration with Azure AD
- Leave management
- Power BI integration
- Automated utilization analytics
- Export to Excel/PDF
- Calendar integration
- AI-based utilization prediction

10. Key Business Rules

Rule	Description
Max Daily Hours	Resource cannot log more than 8 hours/day
Multi-Project Allocation	Allowed
Project Visibility	Only assigned projects visible
Approval Workflow	Mandatory
Weekend Exclusion	Standard working days only
Role-Based Access	Mandatory

11. Expected Deliverables

- Responsive Web Application
- Database Design
- API Documentation
- Source Code Repository
- Deployment Guide
- User Manual
- Test Cases
- Admin Guide

12. Recommended Architecture

A modern 3-tier architecture is recommended:

Presentation Layer

- React/Angular UI

API Layer

- REST APIs (.NET Core/Node.js)

Database Layer

- SQL Server/PostgreSQL

Since you are already working heavily with:

- .NET Core
- Azure
- Microservices
- SQL Server
- Power BI

I would strongly recommend this stack for your organization:

Layer	**Recommended Technology**
Frontend	Angular
Backend	.NET Core Web API
Database	SQL Server
Authentication	Azure AD / JWT
Hosting	Azure App Service
Reporting	Power BI
DevOps	Azure DevOps CI/CD

This will align better with your existing ecosystem and team competency.